

Information constraints in fine-tuning and reasoning language models

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How much information must survive?

Fine-tuning

A compressed update

What must a fixed model receive to retain a learned behavior?

Reasoning

An intermediate state

What must the next computation receive to continue correctly?

Receiver, task, and update rule define what is useful.

Autoregressive language models

$$p_{\theta}(x_1, \dots, x_T) = \prod_t p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$$



Hidden vectors h_t at each layer

Training minimizes next-token negative log-likelihood.

At generation time, each new token extends the prefix.

LoRA and QLoRA

$$W' = W + (\alpha/r) BA$$

Frozen W
4-bit backbone in QLoRA

$d \times d$ values

Trainable A and B
Higher-precision adapter

$2dr$ values, with $r \ll d$

This work compresses the adapter after training.

Coding and rate–distortion

Ideal code length

$$\ell(\mathbf{x}) \approx -\log_2 p(\mathbf{x})$$

Entropy averages this cost.

NLL measures prediction and coding loss.

Lossy coding

minimize rate
subject to acceptable distortion

The receiver already has the base model.
Only the update adds to the file budget.

The distortion measure specifies the behavior to preserve.

Reasoning as sequential computation



Training objectives

Supervised traces
or reward for the final answer

Objects to distinguish

Written tokens and hidden vectors
A state that supports the next step

Decoding a state and using it are separate tests.

An exact rate for a fixed learned update



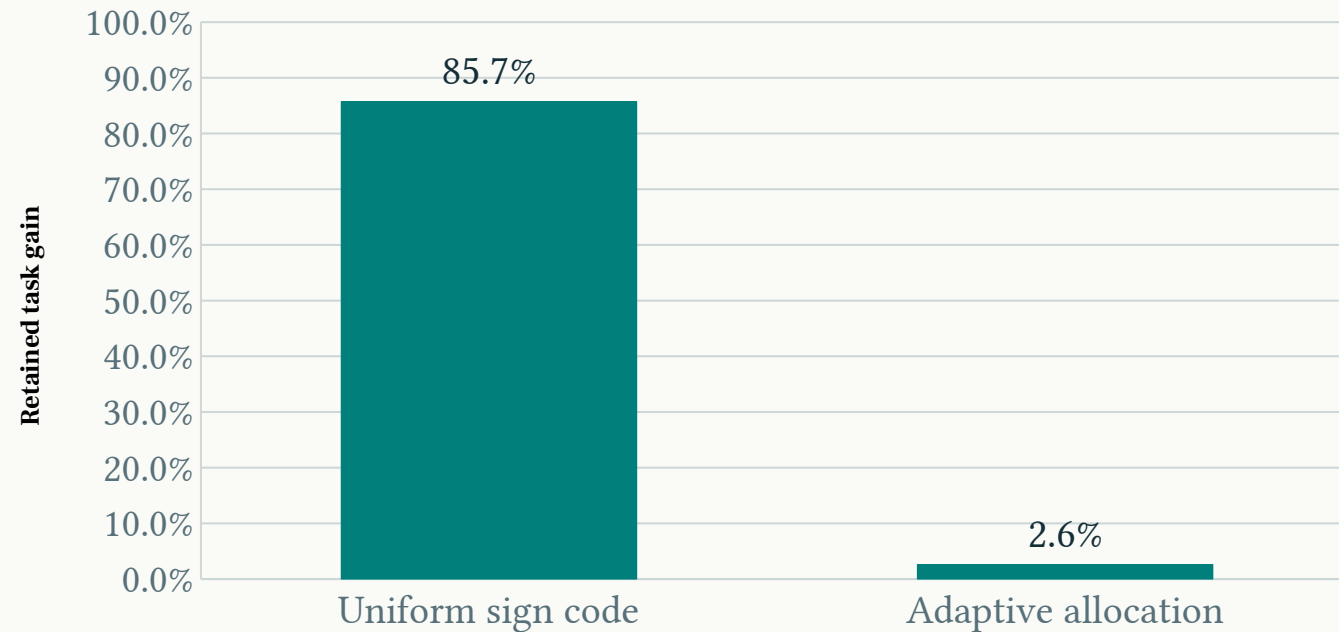
$R = 8 \times \text{file bytes} / \text{original adapter values}$

$\text{Retain}(R) = (L_0 - L_r) / (L_0 - L_{\text{raw}})$

$R^*(0.90)$: smallest measured rate retaining 90% of the gain

L_0 : frozen-model loss L_{raw} : raw-adapter loss L_r : reloaded-adapter loss

Weight error misses behavioral damage



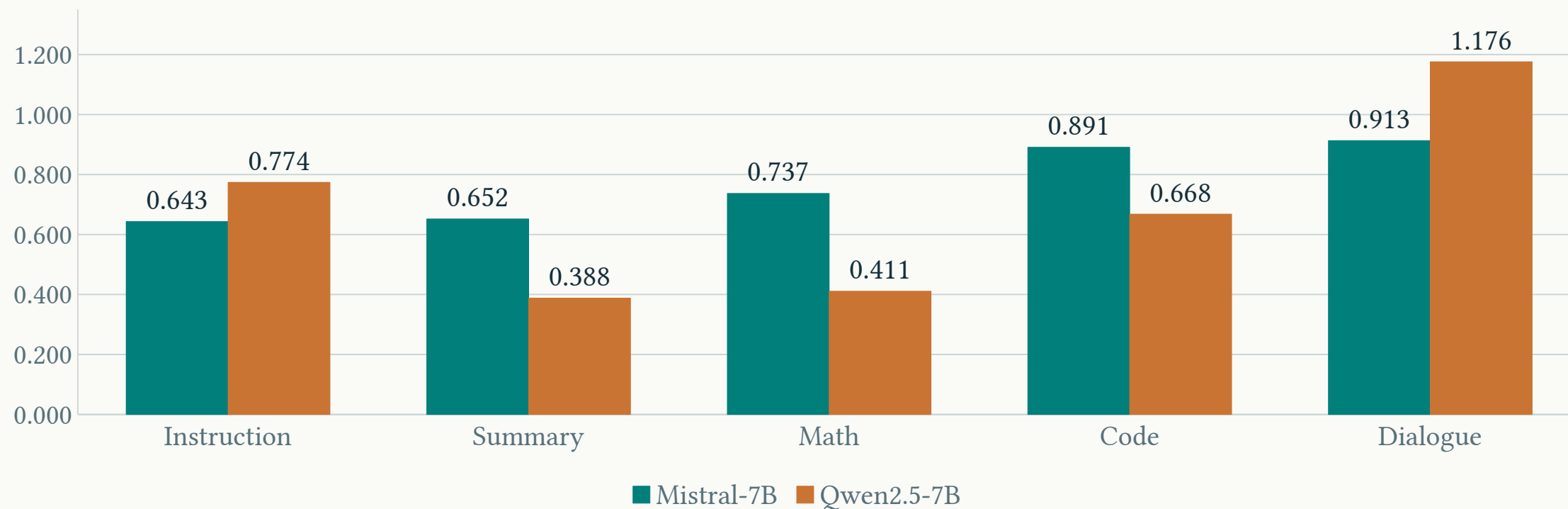
≈ 1 bit / value

Similar weight RMSE
83 points apart in
retained task gain

Behavior must be measured at the model's output.

Mistral-7B + rank-16 LoRA, MetaMathQA. Retained GSM8K exact-match gain, 3 seeds.

The corpus ordering changes with the model



$R^*(0.90)$, bits per original adapter value. Mean over 3 seeds per corpus and model.

Correction spectra are a promising predictor

$$\mathbf{g}_t = \mathbf{h}_t \otimes (\mathbf{p}_t - \mathbf{e}_v)$$

$$\mathbf{F} = \mathbf{G}^\top \mathbf{G} / n$$

$$\mathbf{Icorr} = \log \det(\mathbf{I} \blacklozenge \mathbf{F})$$

Large corrections across many directions

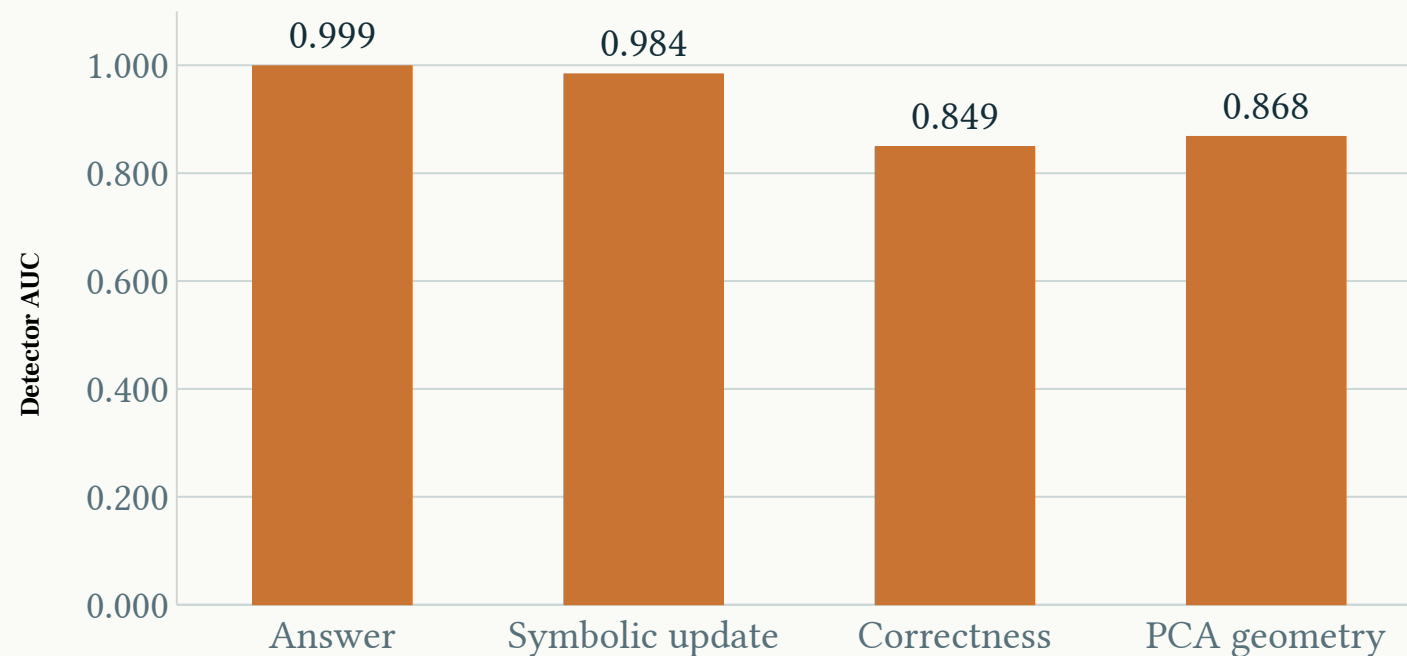
0.764

Within-model Spearman ρ

22 conditions
7 task families
2 frozen 7B models

Projected output-head gradients. Strongest within-model ordering in this screen; prospective validation remains open.

Different objectives select different boundaries



0.098–0.210

Pairwise oracle-boundary F1
within ± 4 tokens

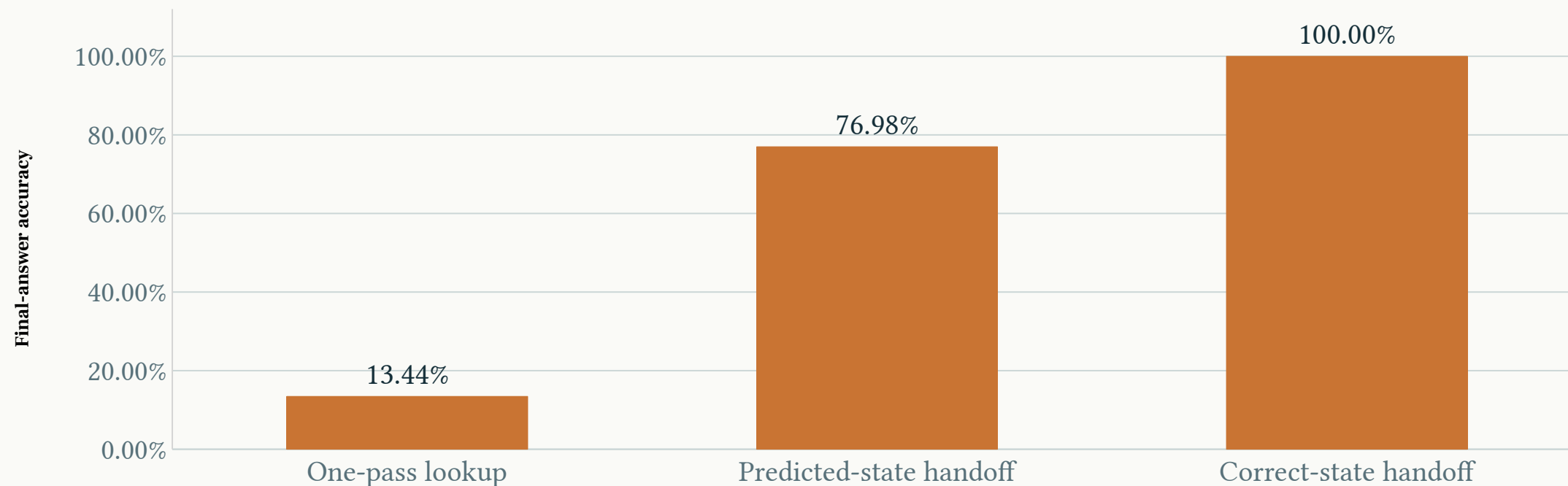
0.020

Best worst-objective utility
among tested rules

SmolLM3-3B. Equal boundary budgets. Random utility = 0, target oracle = 1.

An explicit handoff restores composition

Eight-state program, followed by a state-to-answer lookup



The fresh call receives the state after the history is deleted.

Frozen Qwen2.5-32B, horizon 2, 8 states. Chance = 12.5%. Endpoint inference = 76.98%.

Capacity and stable names matter separately

Code	Bits	Naming	Exact state
Lossy	3	Collisions required	Cannot recover all 16
Canonical	4	One name per state	100% through H = 256
Padded	5	Same stable naming	100% through H = 256
Path-aliased	5	Name depends on path	30–40% at depth 3

At five bits, naming changes state recovery.

16 proof states, matched training data and compute, 3 seeds. H is total horizon; depth counts active deductions.

Local errors accumulate under repeated use

≈ 89%

Correct local transitions

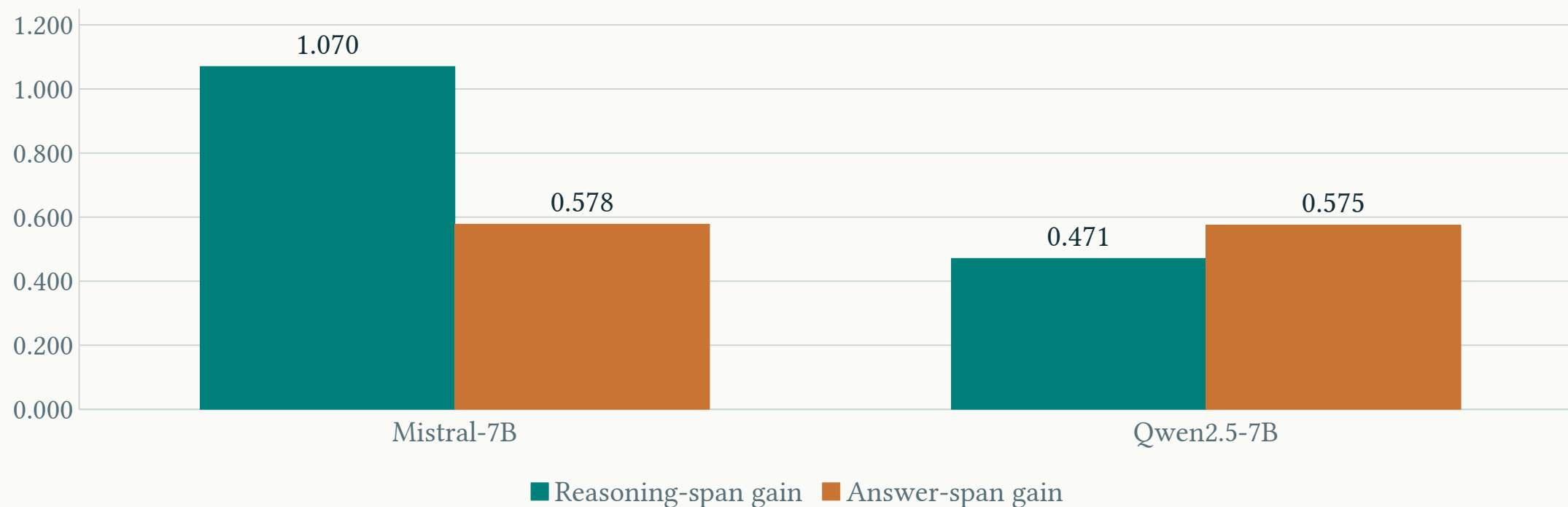
7.92%

Correct final answers at $H = 32$

Chance: 6.25% One-pass reasoning: 5.83%

Reliable single steps do not ensure reliable long executions.

Reasoning and answer rates also reverse



$R^*(0.90)$, bits per adapter value. MetaMathQA. This measures storage for retained next-token gain.

What the experiments establish

Fine-tuning

**A measured file rate
for a stated retained behavior**

The receiver changes the cost.

Reasoning

**A state must support
its intended next computation**

Repeated use adds an update
requirement.

A common interpretation; a predictive law remains open.

The next tests must make new predictions

01 Predict before training

Use span-specific correction spectra on new models and tasks.
Freeze predictions of R^* before observing the adapter rates.

02 Follow state use along the rate ladder

Measure decoding, handoff, and answer behavior
while compressing the same reasoning adapter.

Useful information depends on its use

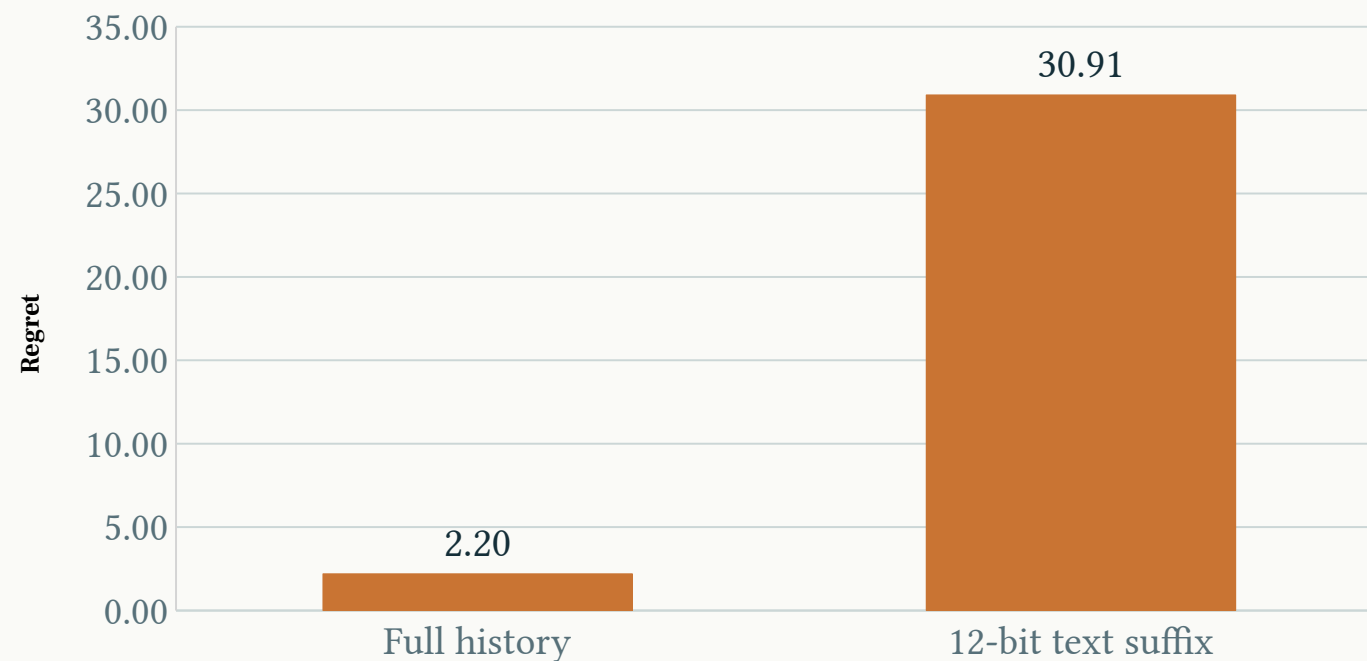
Receiver

Which model must use the information?

Task and update rule

Which behavior must survive, and for how long?

Equal snapshot information, unequal memory



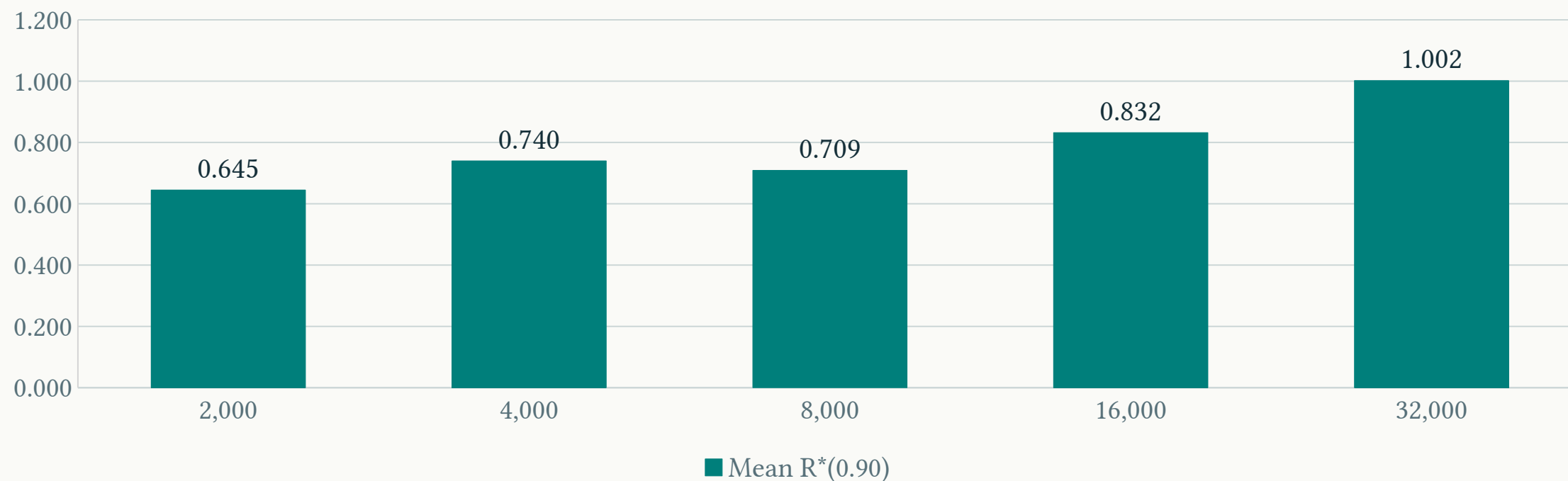
Same initial information

$I(\text{state}; \text{memory}) = 2.512$ bits

Path leakage = 1.781 bits

Exact finite control, 90 task cells. Regret is lower-is-better. This is separate from the LLM assays.

Distinct examples at fixed training exposure



Distinct training examples. 2,000 optimizer steps and 32,000 presentations fixed. 3 seeds per condition.

What the codec and threshold count

Included in the file

Quantized factors
Scales and metadata
Bit-packed, zlib-compressed payload

Conditioned on

A shared frozen backbone
One fixed adapter checkpoint
A named held-out utility

R^* is an attainable rate within the tested codec family.

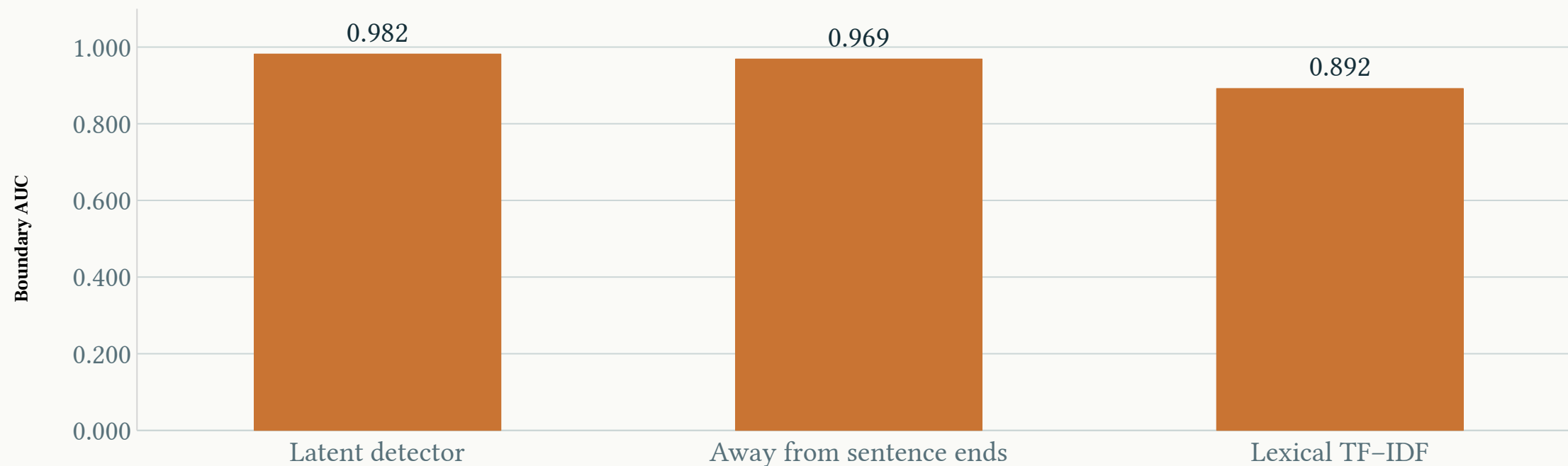
Sub-bit rates use paired rank-direction removal. File rate uses the original adapter value count.

Prediction, accuracy, and state recovery

Metric	Question
Held-out NLL	Does the model predict the scored tokens?
Retained gain	How much improvement over the base survives?
Answer exact match	Is the final task answer correct?
Exact state	Does the complete supplied ledger match?
Boundary AUC / F1	Can we detect a target / do selected gaps agree?

Every reported rate and retention value needs a named utility and evaluation span.

Semantic boundaries are decodable too



1,761 judge-labeled spans, 58 traces. Question-disjoint evaluation. Labels come from Qwen-110B-A25B.

Correction-spectrum screen and its limits

Measure	Within-model ρ	Adjusted p	Held-out RMSE
Correction log-volume	0.764	0.00034	0.187
Correction effective rank	0.609	0.0112	0.181
Surprise-weighted log-volume	0.614	0.0103	0.204
Base target code length	0.200	0.524	0.261
Model-only fit	—	—	0.264

RMSE in bits per adapter value, lower is better. Family-held-out evaluation. 22 conditions, 10 screened measures.